

Vol 8 Chapter 6 — Central Forces and the Kepler Problem

Keeper (author pass — deep math/physics revision)

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Chapter 6 — Central Forces and the Kepler Problem

Level 1 — one sentence

Central forces (force depending only on distance r from a center) reduce the two-body problem to 1D radial motion with effective potential $V_{\text{eff}}(r) = V(r) + L^2/(2mr^2)$, and the inverse-square Kepler-Coulomb case has an “accidental” $SO(4)$ symmetry (Runge-Lenz vector conserved) that’s actually inherited from the substrate $SO(5) \supset SO(4) \supset SO(3)$ chain — Vol 5 Ch 6 hydrogen reuses the same structure.

Level 2 — graduate-physicist precision

6.1 The two-body problem reduction

Two bodies interacting through central potential $V(|\vec{r}_1 - \vec{r}_2|)$. Define: - Center of mass: $\vec{R} = (m_1\vec{r}_1 + m_2\vec{r}_2)/M$, $M = m_1 + m_2$ - Relative position: $\vec{r} = \vec{r}_2 - \vec{r}_1$ - Reduced mass: $\mu = m_1 m_2 / M$

The motion decouples: $\vec{R}$ moves uniformly (no external force), and $\vec{r}$ obeys

$$\mu \ddot{\vec{r}} = -\nabla V(r)$$

This is a single-body problem with mass μ in central potential $V(r)$.

6.2 Conservation laws

Central force is rotation-invariant $\rightarrow$ angular momentum $\vec{L} = \mu \vec{r} \times \dot{\vec{r}}$ conserved (Ch 5).

Motion is planar (perpendicular to $\vec{L}$). In the plane, use polar coordinates (r, ϕ) .

Energy $E = (1/2)\mu(\dot{r}^2 + r^2\dot{\phi}^2) + V(r)$ conserved.

From angular momentum: $L = \mu r^2 \dot{\phi} \rightarrow \dot{\phi} = L/(\mu r^2)$. Substituting:

$$E = \frac{1}{2}\mu\dot{r}^2 + \frac{L^2}{2\mu r^2} + V(r) = \frac{1}{2}\mu\dot{r}^2 + V_{\text{eff}}(r)$$

with effective potential $V_{\text{eff}}(r) = V(r) + L^2/(2\mu r^2)$ — the second term is the centrifugal barrier.

6.3 The Kepler problem: inverse-square attractive force

Gravity: $V(r) = -GMm/r = -k/r$ (with $k = GMm$).

Effective potential: $V_{\text{eff}}(r) = -k/r + L^2/(2\mu r^2)$.

Equation of orbit ($u = 1/r$):

$$\frac{d^2u}{d\phi^2} + u = \mu k/L^2$$

Solution: $u = (1 + e \cos \phi)/p$ with $p = L^2/(\mu k)$ and e eccentricity. This is a conic section in polar coordinates: - $e = 0$: circle - $0 < e < 1$: ellipse (bound orbit) - $e = 1$: parabola (just escapes) - $e > 1$: hyperbola (unbound)

6.4 Kepler's three laws

From the orbit equation:

Kepler's first law: planets move in ellipses with the sun at one focus.

Kepler's second law: line from sun to planet sweeps equal areas in equal times. Direct consequence of angular momentum conservation: $dA/dt = L/(2\mu) = \text{constant}$.

Kepler's third law: $T^2 \propto a^3$ where a is semi-major axis. Specifically $T^2 = 4\pi^2\mu a^3/k$.

These are the famous laws Kepler extracted from Tycho Brahe's data (1609, 1619) — predating Newton's gravity by 70+ years.

6.5 The Runge-Lenz vector and hidden $SO(4)$ symmetry

The Kepler problem has an additional conserved quantity beyond $\vec{L}$ and E :

$$\vec{A} = \vec{p} \times \vec{L} - \mu k \hat{r}$$

This is the Runge-Lenz vector (Laplace, Runge, Lenz). Magnitude $|\vec{A}| = \mu k e$.

The conservation of $\vec{A}$ is special to the $1/r$ potential — it's not present for general $V(r)$. Together with $\vec{L}$, the Runge-Lenz vector forms an $SO(4)$ Lie algebra structure, expanding the obvious $SO(3)$ symmetry.

6.6 Substrate origin of $SO(4)$: the $SO(5) \supset SO(4)$ chain

Volume 5 Chapter 6 Section 6.2-6.3: in BST, the Coulomb-Kepler problem's hidden $SO(4)$ is inherited from the substrate $SO(5) \supset SO(4) \supset SO(3)$ chain in the maximal compact subgroup $K = SO(5) \times SO(2)$.

The substrate-mechanism reading: inverse-square central forces happen to respect a substrate-natural subgroup $SO(4)$ that's NOT present for other central forces. The Kepler "accidental" symmetry is not accidental — it's the substrate's $SO(5)$ structure restricted to the inverse-square sector.

This is the same mechanism that gives hydrogen its n^2 degeneracy in quantum mechanics (Volume 5 Chapter 6). The classical and quantum versions of the Kepler problem share the same substrate hidden symmetry.

6.7 Other central forces

- Harmonic ($V = (1/2)kr^2$): orbits are closed ellipses centered at origin (Bertrand's theorem)
- Power-law ($V = -k/r^n$, $n \neq 1, -2$): orbits are not closed (precess)
- Yukawa ($V = -ke^{-\mu r}/r$): partially closed; nuclear-force model

Bertrand's theorem (1873): only Kepler ($V = -k/r$) and harmonic ($V = kr^2/2$) give closed orbits for all bounded motions. Both have hidden symmetries: Kepler $SO(4)$, harmonic $SU(3)$.

6.8 Worked example: Earth's orbit

Earth-Sun: - $a = 1.496 \times 10^{11} \text{ m} = 1 \text{ AU}$ (semi-major axis) - $e \approx 0.0167$ (nearly circular) - $T = 365.25 \text{ days} = 3.156 \times 10^7 \text{ s}$ - Earth mass $m \approx 5.97 \times 10^{24} \text{ kg}$ - Sun mass $M \approx 1.99 \times 10^{30} \text{ kg}$

Kepler's third law check: $T^2 = 4\pi^2 a^3 / (GM)$. $GM \approx 1.327 \times 10^{20}$. Compute: $4\pi^2 a^3 / (GM) = 39.48 \cdot (1.496 \times 10^{11})^3 / (1.327 \times 10^{20}) \approx 9.96 \times 10^{14}$. $\sqrt{} \approx 3.156 \times 10^7 \text{ s}$. Matches.

6.9 K-audit anchors

- Volume 5 Chapter 6: hydrogen atom (quantum analog with same $SO(4)$ symmetry)
- Volume 5 Chapter 3: $SO(5)$ substrate angular momentum (parent group)
- Substrate $SO(5) \supset SO(4) \supset SO(3)$ chain: substrate origin of Kepler symmetry

Level 3 — 5th-grader accessibility

Central forces depend only on distance, not direction. Examples: gravity (inverse-square pull), spring (linear pull), nuclear forces. The two-body problem reduces to 1D radial motion plus angular conservation. Kepler's three laws: 1. Planets move in ellipses with sun at one focus 2. Equal areas swept in equal times (= angular momentum conservation) 3. $T^2 \propto a^3$ (orbital period squared = semi-major axis cubed)

These laws describe all gravitational orbits (planets, satellites, binaries). They predate Newton by decades! The Kepler problem has an extra "accidental" symmetry beyond rotation — the Runge-Lenz vector — that makes orbits close into perfect ellipses (other central forces don't have this; their orbits precess). In BST, this isn't accidental: it's the substrate's $SO(5)$ structure showing up when the potential happens to be $1/r$. Same reason hydrogen orbitals have n^2 degeneracy (Vol 5 Ch 6).

What comes next

Chapter 7 develops rigid body dynamics.

Where to look this up

- Kepler 1609, 1619 — original three laws
- Newton 1687 — gravity derivation
- Goldstein Ch 3; Marion-Thornton Ch 8-9
- BST: Volume 5 Chapter 6 (quantum hydrogen analog)